

Language Translation Assistant

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SECTION: E

PROBLEM STATEMENT:

Implementing a utility-based NLP agent to build a Language Translation Assistant that translates user manuals from English to other languages such as French and Spanish. The system takes a user manual as an input file and generates a translated versions using NLP models.

Abstract

Many products come with user manuals written only in English, which can be difficult for non-English users to understand. In this project, a Language Translation Assistant is developed to translate English user manuals into French and Spanish. The system uses Transformer-based models to perform the translation. The manual is given as an input file, and the translated content is generated as output files. This project shows how NLP models can be used for real-world document translation.

Observation

Translation is a Natural Language Processing task where one language is converted into another. Encoder-only models like BERT are mainly used for understanding text and are not suitable for translation. Encoder–Decoder models are better for translation because they can generate new text.

A user manual usually means a document that contains instructions, warnings, and steps.

Built a Language Translation Assistant using Python.

The system takes a user manual text file as input.

The content of the file is translated into French and Spanish.

The translated output is saved into separate files for each language.

The project is implemented using Hugging Face Transformer models in a Jupyter Notebook.

Tools Used

Python

Hugging Face Transformers

MarianMT translation models

Jupyter Notebook

Output screenshots:

```
Translating text

In [ ]: text = """
This device should be connected to a power source before use.
Ensure that the power button is pressed for three seconds.
"""

In [ ]: #Translating to french
french_output = translator_fr(text, max_length=400)
print("French Translation:\n", french_output[0]['translation_text'])

French Translation:
Ce dispositif doit être connecté à une source d'alimentation avant utilisation. Assurez-vous que le bouton d'alimentation est appuyé pendant trois secondes.

In [ ]: #Translating to spanish
spanish_output = translator_es(text, max_length=400)
print("\nSpanish Translation:\n", spanish_output[0]['translation_text'])

Spanish Translation:
Este dispositivo debe estar conectado a una fuente de alimentación antes de su uso. Asegúrese de que el botón de encendido se presiona durante tres segundos.
```

Reading user manual as input document

```
def read_manual(file_path):
    with open(file_path, "r", encoding="utf-8") as f:
        return f.read()

manual_text = read_manual("input_eng.txt")
print(manual_text)

Read all safety instructions carefully before operating the product.
Keep out of reach of children.
Do not attempt to disassemble the device.
Stop using the product if it becomes damaged.
Plug the appliance into a grounded power outlet.
Press the start button to begin operation.
Allow the appliance to cool down before cleaning.
Unplug the appliance after use.
Create an account to access all features.
Do not share your password with others.
Update the application regularly for best performance.
Contact support if you experience any issues.

In [ ]: french_manual = translate_manual(manual_text, translator_fr)
print("French Translation:\n", french_manual)

French Translation:
Lisez attentivement toutes les instructions de sécurité avant d'utiliser le produit. Restez hors de portée des enfants. N'essayez pas de démonter l'appareil. Arrêtez d'utiliser le produit s'il est endommagé. Branchez l'appareil sur une prise d'alimentation à la terre. Appuyez sur le bouton de démarrage pour commencer à fonctionner. Laissez l'appareil refroidir avant de nettoyer. Débranchez l'appareil après utilisation. Créez un compte pour accéder à toutes les fonctionnalités. Ne partagez pas votre mot de passe avec d'autres. Mettez l'application à jour régulièrement pour obtenir les meilleures performances.

In [ ]: spanish_manual = translate_manual(manual_text, translator_es)
```

```
: spanish_manual = translate_manual(manual_text, translator_es)
print("\nSpanish Translation:\n", spanish_manual)
```

Spanish Translation:

Lea cuidadosamente todas las instrucciones de seguridad antes de operar el producto. Manténgase fuera del alcance de los niños. No intente desmontar el dispositivo. Detenga el uso del producto si se daña. Conecte el aparato a una toma de corriente a tierra. Pulse el botón de inicio para iniciar la operación. Permita que el aparato se enfríe antes de limpiar. Desenchufe el aparato después de su uso. Cree una cuenta para acceder a todas las funciones. No comparta su contraseña con otros. Actualice la aplicación regularmente para obtener el mejor rendimiento. Comuníquese con el soporte técnico si experimenta algún problema.

Writing the translated text

```
: with open("manual_french.txt", "w", encoding="utf-8") as f:
    f.write(french_manual)

with open("manual_spanish.txt", "w", encoding="utf-8") as f:
    f.write(spanish_manual)
```

 jupyter manual_spanish.txt Last Checkpoint: 13 minutes ago

File Edit View Settings Help

```
1 | Lea cuidadosamente todas las instrucciones de seguridad antes de operar el producto. Manténgase fuera del alcance de los niños. No intente desmontar el
    dispositivo. Detenga el uso del producto si se daña. Conecte el aparato a una toma de corriente a tierra. Pulse el botón de inicio para iniciar la operación.
    Permita que el aparato se enfríe antes de limpiar. Desenchufe el aparato después de su uso. Cree una cuenta para acceder a todas las funciones. No comparta su
    contraseña con otros. Actualice la aplicación regularmente para obtener el mejor rendimiento. Comuníquese con el soporte técnico si experimenta algún problema.
```

 jupyter manual_french.txt Last Checkpoint: 14 minutes ago

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```
1 | Lisez attentivement toutes les instructions de sécurité avant d'utiliser le produit. Restez hors de portée des enfants. N'essayez pas de démonter l'appareil.
    Arrêtez d'utiliser le produit s'il est endommagé. Branchez l'appareil sur une prise d'alimentation à la terre. Appuyez sur le bouton de démarrage pour commencer à
    fonctionner. Laissez l'appareil refroidir avant de nettoyer. Débranchez l'appareil après utilisation. Créez un compte pour accéder à toutes les fonctionnalités.
    Ne partagez pas votre mot de passe avec d'autres. Mettez l'application à jour régulièrement pour obtenir les meilleures performances.
```